EXPERIMENT - 2

AIM: To perform Exploratory Data Analysis (EDA) and data preprocessing in Python using Pandas, including data cleaning, handling missing values, normalization, and encoding for analytics.

```
import pandas as pd

url = r"C:/Users/shrey  patel/Downloads/titanic.csv"
df = pd.read_csv(url)

print("Dataset loaded successfully!")
```

Dataset loaded successfully!

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
print("First 5 rows:")
print(df.head())
```

```
First 5 rows:
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

```
print("\nLast 5 rows:")
print(df.tail())
```

```
Last 5 rows:
     PassengerId  Survived  Pclass                                      Name  \
886          887         0       2                     Montvila, Rev. Juozas
887          888         1       1              Graham, Miss. Margaret Edith
888          889         0       3  Johnston, Miss. Catherine Helen "Carrie"
889          890         1       1                     Behr, Mr. Karl Howell
890          891         0       3                       Dooley, Mr. Patrick

        Sex   Age  SibSp  Parch      Ticket   Fare Cabin Embarked
886    male  27.0      0      0      211536  13.00   NaN        S
887  female  19.0      0      0      112053  30.00   B42        S
888  female   NaN      1      2  W./C. 6607  23.45   NaN        S
889    male  26.0      0      0      111369  30.00  C148        C
890    male  32.0      0      0      370376   7.75   NaN        Q
```

```
print("\nDataset shape:", df.shape)
```

Dataset shape: (891, 12)

```
print("\nDataset info:")
print(df.info())
```

Dataset info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):

Column | Non-Null Count | Data Type

0 PassengerId 891 non-null int64

1 Survived 891 non-null int64

2 Pclass 891 non-null int64

3 Name 891 non-null object

4 Sex 891 non-null object

5 Age 714 non-null float64

6 SibSp 891 non-null int64

7 Parch 891 non-null int64

8 Ticket 891 non-null object

9 Fare 891 non-null float64

10 Cabin 204 non-null object

11 Embarked 889 non-null object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None

```
print("\nSummary statistics:")
print(df.describe())
```

```
Summary statistics:
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

```
print("\nMissing values per column:")
print(df.isnull().sum())
```

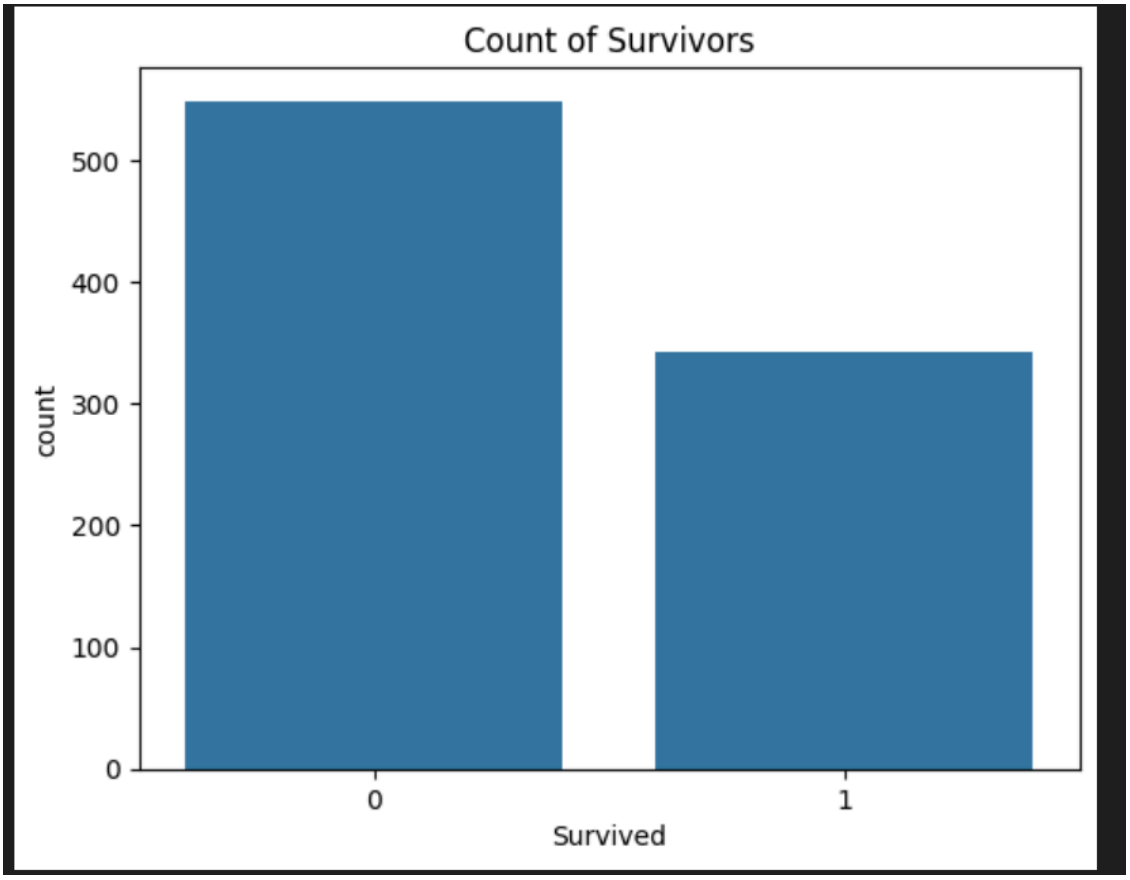
```
Missing values per column:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
print("\nUnique values per column:")
for col in df.columns:
```

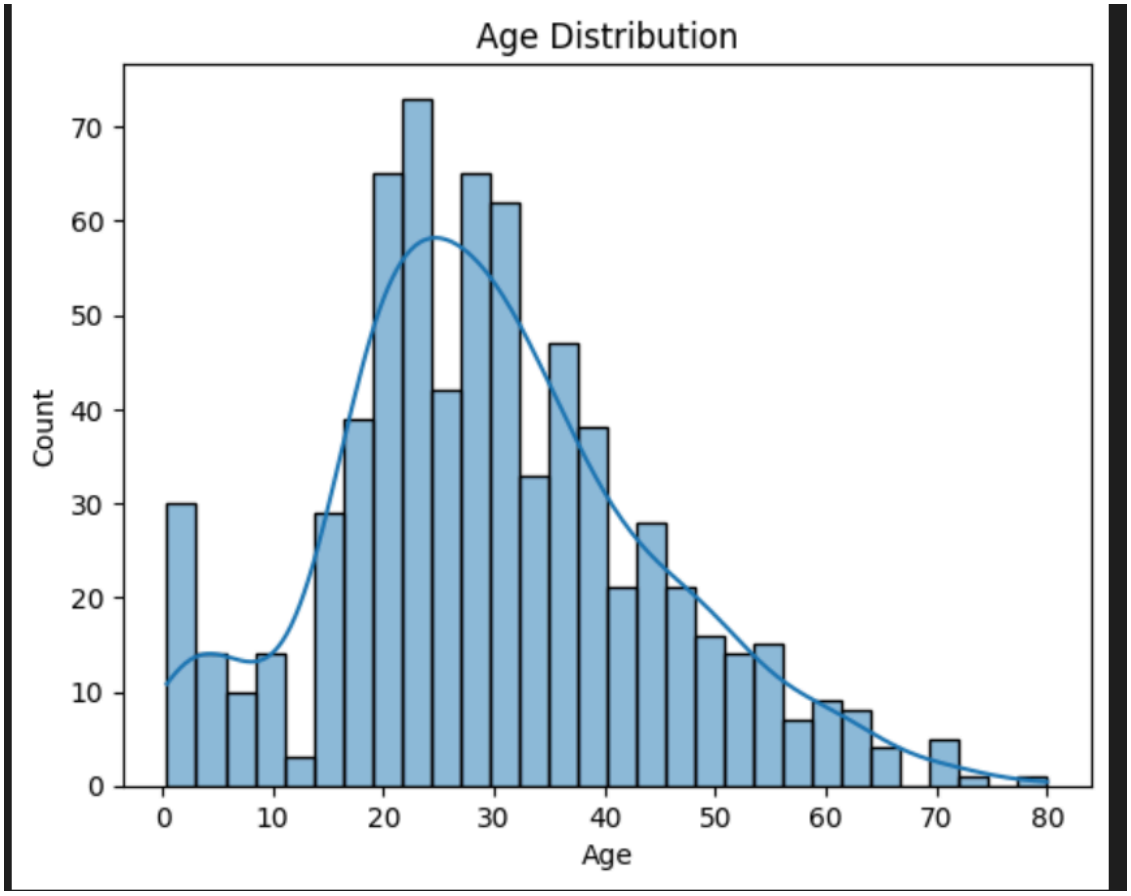
```
    print(f"{col}: {df[col].nunique()}")
```

Unique values per column:
PassengerId: 891
Survived: 2
Pclass: 3
Name: 891
Sex: 2
Age: 88
SibSp: 7
Parch: 7
Ticket: 681
Fare: 248
Cabin: 147
Embarked: 3

```
import matplotlib.pyplot as plt
import seaborn as sns
sns.countplot(x='Survived', data=df)
plt.title("Count of Survivors")
plt.show()
```




```
import matplotlib.pyplot as plt
import seaborn as sns
sns.histplot(df['Age'], bins=30, kde=True)
plt.show()
```




```
df['Age'].fillna(df['Age'].median(), inplace=True)
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=Tru
e)
print("\nHandled missing values.")
```

Handled missing values.

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked
0	0	3	male	22.0	1	0	7.2500	S
1	1	1	female	38.0	1	0	71.2833	C
2	1	3	female	26.0	0	0	7.9250	S
3	1	1	female	35.0	1	0	53.1000	S
4	0	3	male	35.0	0	0	8.0500	S

```
print("\nProcessed dataset info:")
print(df.info())
print(df.head())
```

Processed dataset info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 8 columns):

Column Non-Null Count Dtype

---

0 Survived 891 non-null int64
1 Pclass 891 non-null int64
2 Sex 891 non-null int64
3 Age 891 non-null float64
4 SibSp 891 non-null float64
5 Parch 891 non-null float64
6 Fare 891 non-null float64
7 Embarked 891 non-null int64
dtypes: float64(4), int64(4)
memory usage: 55.8 KB
None

```
X = df.drop('Survived', axis=1)
y = df['Survived']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
print("\nTrain/Test split done.")
```

Train/Test split done.

Image Dataset

```
import numpy as np
import matplotlib.pyplot as plt
import cv2
```

```
from tensorflow.keras.datasets import cifar10
from tensorflow.keras.preprocessing.image import ImageDataG
enerator


# 1. Read any inbuilt image dataset.(RGB image)

(x_train, y_train), (x_test, y_test) = cifar10.load_data()

print("Dataset loaded successfully")
```

Downloading data from

https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz

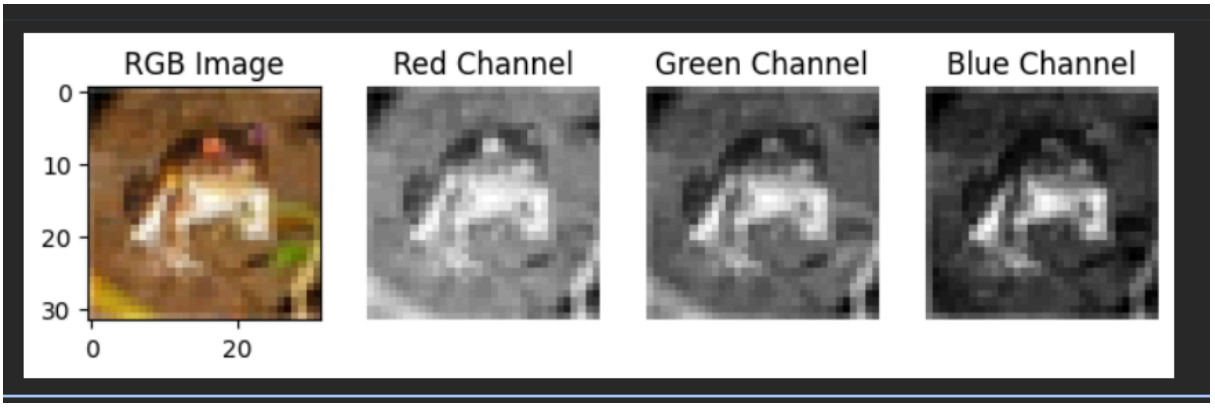
170498071/170498071 ━━━━ 3s
Dataset loaded successfully

```
print("\nDataset Information:")
print("Training images shape :", x_train.shape)
print("Training labels shape :", y_train.shape)
print("Test images shape     :", x_test.shape)
print("Single image shape    :", x_train[0].shape)
```

Dataset Information:
Training images shape : (50000, 32, 32, 3)
Training labels shape : (50000, 1)
Test images shape : (10000, 32, 32, 3)
Single image shape : (32, 32, 3)

```
img_rgb = x_train[0]

plt.figure(figsize=(8, 4))

plt.subplot(1, 4, 1)
plt.imshow(img_rgb)
plt.title("RGB Image")
plt.axis("on")
```

```
plt.subplot(1, 4, 2)
plt.imshow(img_rgb[:, :, 0], cmap="gray")
plt.title("Red Channel")
plt.axis("off")

plt.subplot(1, 4, 3)
plt.imshow(img_rgb[:, :, 1], cmap="gray")
plt.title("Green Channel")
plt.axis("off")

plt.subplot(1, 4, 4)
plt.imshow(img_rgb[:, :, 2], cmap="gray")
plt.title("Blue Channel")
plt.axis("off")

plt.show()
```



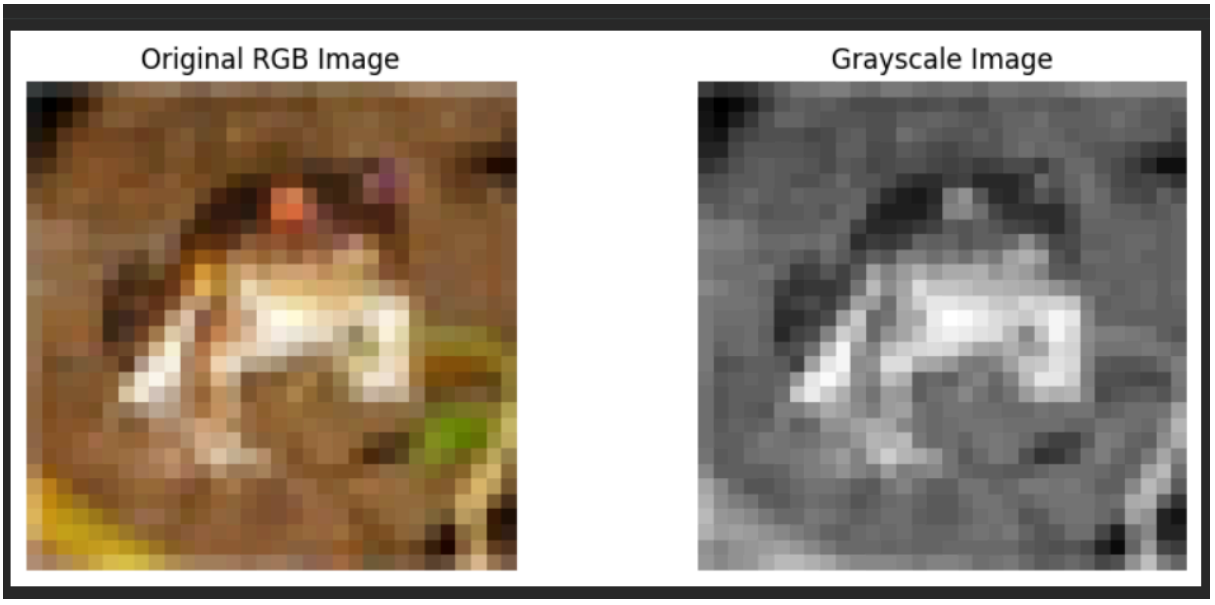

```
print("\nRGB Pixel Values (Top-left 5x5 pixels):")
print(img_rgb[0:5, 0:5, :])
```

RGB Pixel Values (Top-left 5×5 pixels):
[[[59 62 63]
[43 46 45]
[50 48 43]

```
[ 68  54  42]
[ 98  73  52]]

[[ 16  20  20]
[  0   0   0]
[ 18   8   0]
[ 51  27   8]
[ 88  51  21]]

[[ 25  24  21]
[ 16   7   0]
[ 49  27   8]
[ 83  50  23]
[110  72  41]]

[[ 33  25  17]
[ 38  20   4]
[ 87  54  25]
[106  63  28]
[115  70  33]]

[[ 50  32  21]
[ 59  32  11]
[102  65  34]
[127  79  39]
[124  77  36]]]
```

```
img_gray = cv2.cvtColor(img_rgb, cv2.COLOR_RGB2GRAY)

plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.imshow(img_rgb)
plt.title("Original RGB Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(img_gray, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")
```

```
plt.show()
```




```
img_normalized = img_rgb / 255.0

print("\nRGB Pixel Values (Top-left 4x5 pixels):")
print(img_rgb[0:4, 0:5])

print("\n Normalized image Pixel Values (Top-left 4x5 pixel
s):")
print(img_normalized[0:4, 0:5])
```

Normalized image Pixel Values (Top-left 4×5 pixels):
[[[0.23137255 0.24313725 0.24705882]
[0.16862745 0.18039216 0.17647059]
[0.19607843 0.18823529 0.16862745]
[0.26666667 0.21176471 0.16470588]
[0.38431373 0.28627451 0.20392157]]

[[0.0627451 0.07843137 0.07843137]
[0.   0.   0.   ]
[0.07058824 0.03137255 0.   ]

```
[0.2        0.10588235 0.03137255]
[0.34509804 0.2        0.08235294]]

[[0.09803922 0.09411765 0.08235294]
[0.0627451  0.02745098 0.        ]
[0.19215686 0.10588235 0.03137255]
[0.3254902  0.19607843 0.09019608]
[0.43137255 0.28235294 0.16078431]]

[[0.12941176 0.09803922 0.06666667]
[0.14901961 0.07843137 0.01568627]
[0.34117647 0.21176471 0.09803922]
[0.41568627 0.24705882 0.10980392]
[0.45098039 0.2745098  0.12941176]]]
```